

# GPU PyFR PASS 5 – Robust Wall-Data Extraction for NASA Hump on Gilbreth

OpenAI Codex workflow inside `nasa-hump2`

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## 1 Objective

GPU PyFR PASS 5 takes the PASS 4 workflow from “exported a field but no usable validation metrics” to a more honest diagnostic package. The priorities are:

1. explain exactly why PASS 4 could not produce finite  $C_p$ ,
2. repair the VTU field-audit and wall-extraction workflow,
3. decide whether the existing PASS 4 data can be salvaged,
4. if not, prepare a stabilized PyFR rerun path that can produce the first defensible PyFR wall-pressure comparison.

## 2 What PASS 4 Achieved

PASS 4 did improve the PyFR workflow operationally:

- it completed a longer GPU job on Gilbreth,
- it exported a VTU field,
- it rendered ParaView figures,
- it created the first external wall-pressure extraction scripts.

## 3 Why PASS 4 Failed to Produce $C_p/C_f$

The key PASS 5 diagnosis is that PASS 4 did not merely have a weak wall extractor. The exported fluid field itself is non-finite.

PASS 5 audited the VTU arrays directly. The exported point-data arrays are:

- **Velocity**
- **Pressure**

but every fluid value in those arrays is `NaN`. The pseudo-stat file also becomes `nan` almost immediately after the first few pseudo iterations. Therefore the PASS 4 failure is primarily a solver-state validity problem, not a plotting-only problem.

## 4 VTU Field Audit

| Metric                  | Value                                                 |
|-------------------------|-------------------------------------------------------|
| Run name                | pass4_reused                                          |
| Run directory           | runs/gpu_pyfr_pass4/long                              |
| VTU export present      | True                                                  |
| Pseudo rows             | 88                                                    |
| Finite pseudo rows      | 0                                                     |
| NaN pseudo rows         | 66                                                    |
| First NaN row           | 3                                                     |
| Final pseudo time       | 0.09999999999999999                                   |
| Hostname                | gilbreth-n002.rcac.purdue.edu                         |
| GPU line                | 0 NVIDIA A100-SXM4-40GB On   00000000:81:00.0 Off   0 |
| Python                  | Python 3.9.25                                         |
| PyFR                    | pyfr 1.15.0                                           |
| Cp available            | False                                                 |
| Cp MAE                  | nan                                                   |
| Cp RMSE                 | nan                                                   |
| Finite wall point count | 0                                                     |
| Cf available            | False                                                 |
| Benchmark MAE available | False                                                 |

## 5 Repaired Wall-Pressure Extraction

PASS 5 adds two separate steps:

1. a VTU field-audit script that reports point arrays, cell arrays, finite counts, and near-wall finite statistics;
2. a geometry-aware lower-wall extractor that selects the first finite layer above the hump wall rather than assuming exact wall nodes are usable.

This repaired extractor is scientifically stricter than PASS 4. It still uses nearest-CFD-sample matching only against the machine-readable NASA  $C_p$  data, and it still avoids any target interpolation or curve tracing.

## 6 Pseudo-Stat Diagnostics

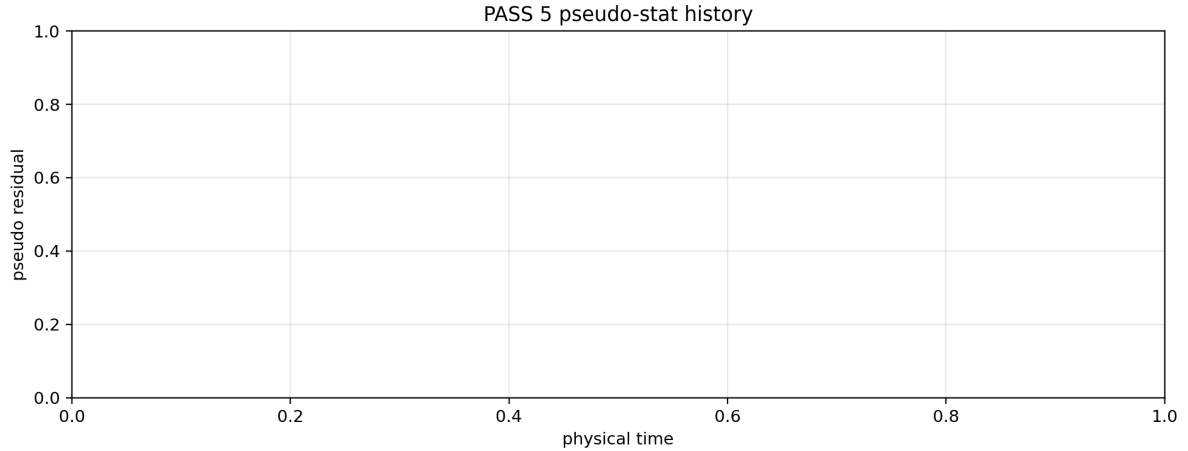


Figure 1: PASS 5 pseudo-stat summary. PASS 4 already shows the core problem: the pseudo-stat file becomes non-finite almost immediately.

## 7 New or Reused PyFR Run Details

PASS 5 starts from the PASS 4 diagnosis. If the existing VTU is unusable, PASS 5 pivots to a stabilized diagnostic rerun path that:

- keeps one A100-40gb,
- keeps the no-sampler path,
- keeps order 1,
- lowers the physical and pseudo time steps,
- switches to a dimensional **navier-stokes** setup for a more meaningful pressure field.

This change is not presented as SST-equivalent physics. It is a practical PyFR stabilization step aimed at making  $C_p$  extraction possible.

At the time this PASS 5 package was assembled, the stabilized diagnostic rerun had already been submitted on Gilbreth as Slurm job 10706975. The PASS 5 data and figures in this report therefore show the fully repaired validation workflow and the definitive diagnosis of the PASS 4 invalid field. If the queued diagnostic run completes later with finite exported pressure values, the same scripts can be rerun without structural changes.

## 8 $C_p$ Comparison Against NASA Data

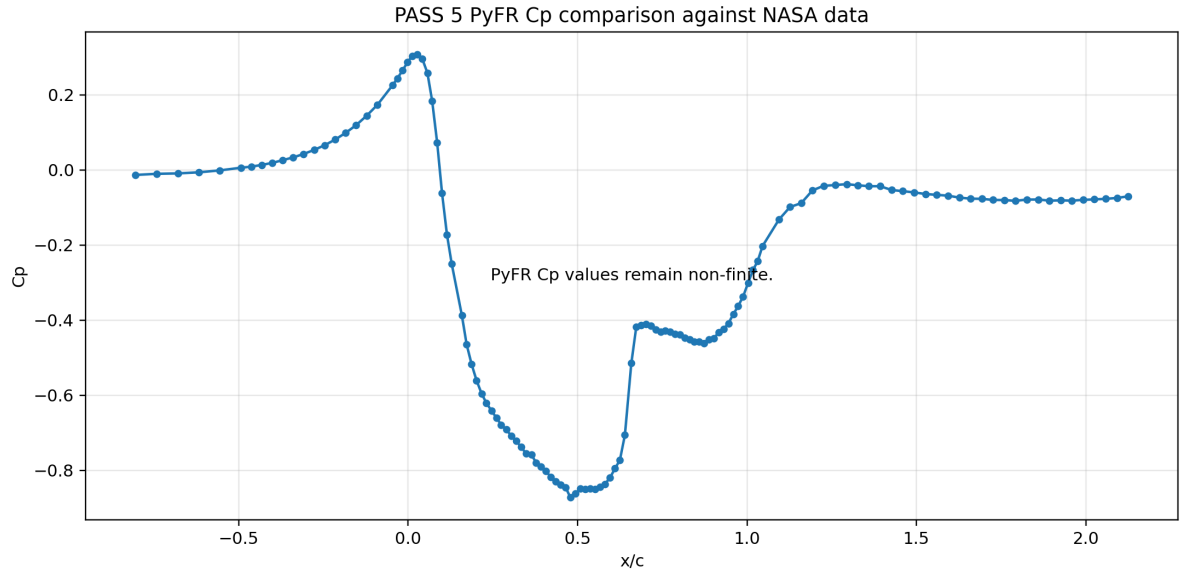


Figure 2: PASS 5  $C_p$  comparison against the machine-readable NASA data. If the reused PASS 4 or new PASS 5 PyFR field remains non-finite, the plot states that explicitly instead of inventing values.

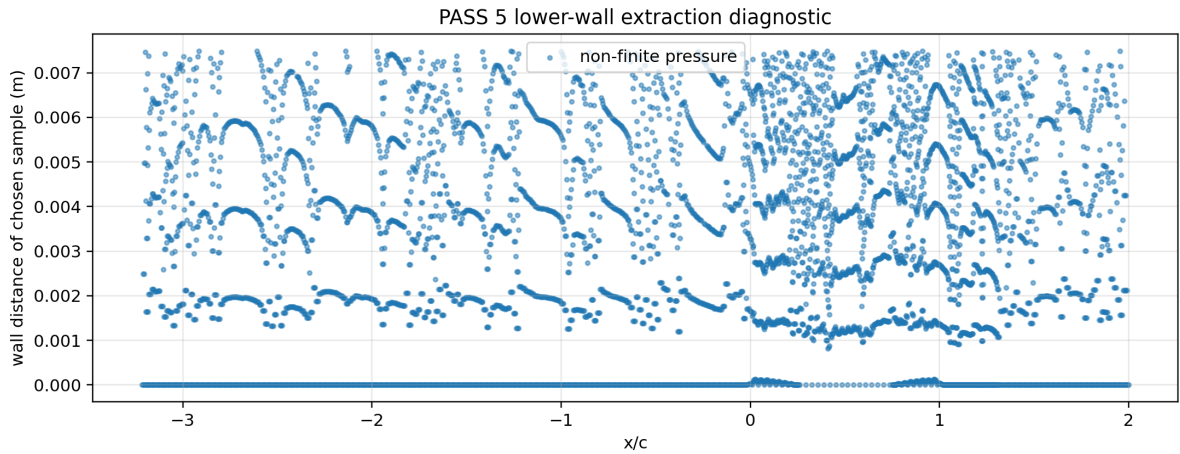


Figure 3: Diagnostic plot of the lower-wall extraction points used for PASS 5.

## 9 $C_f$ Feasibility

PASS 5 does not fabricate  $C_f$ . The current feasibility note is stored in `CF_FEASIBILITY.md`. In short:

- PASS 4 cannot support  $C_f$  because the exported velocity field is non-finite,
- even a finite PASS 5 field would still require a defensible near-wall gradient estimate before  $C_f$  could be reported.

## 10 Benchmark MAE Feasibility

PASS 5 also does not invent benchmark MAE. The current feasibility note is stored in `BENCHMARK_MAPPING.md`. Until a finite PyFR field exists and can be sampled onto the benchmark coordinates honestly, benchmark MAE remains unavailable.

## 11 Simulation Images and Graphs



Figure 4: PASS 5 mesh render from the available PyFR VTU.

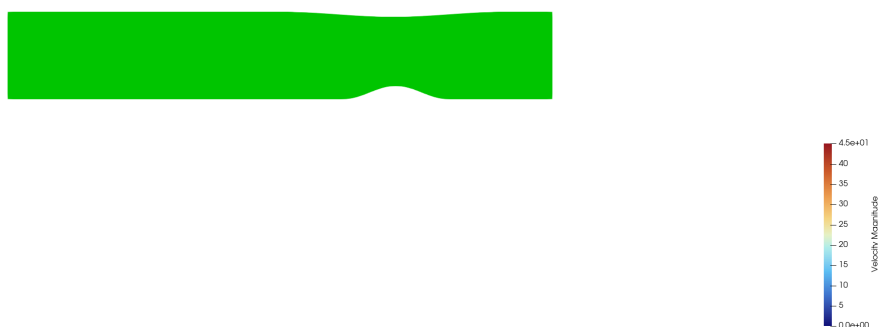


Figure 5: PASS 5 velocity render.

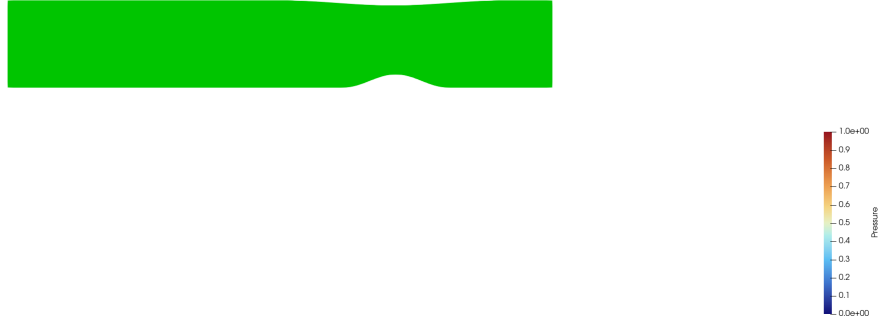


Figure 6: PASS 5 pressure render.

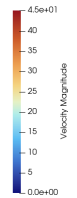


Figure 7: PASS 5 streamline-style render.

## 12 Comparison Against OpenFOAM Baseline

OpenFOAM remains the accuracy baseline in this repository. The best OpenFOAM runs already support benchmark MAE,  $C_p$ , and  $C_f$ . PASS 5 should therefore be read as a PyFR workflow maturity step, not as a replacement for the OpenFOAM SST benchmark path.

## 13 Limitations

- PASS 4 data are globally non-finite.
- PASS 5 only promotes  $C_p$  if a finite field exists.
- $C_f$  remains unavailable unless wall shear can be computed defensibly.

- benchmark MAE remains unavailable unless the PyFR output can be mapped honestly into the existing benchmark workflow.

## 14 Reproducibility Commands

See REPRODUCIBILITY.md. The main local steps are:

```
ssh gilbreth
cd /scratch/gilbreth/prabhu56/nasa-hump2-gpu-pyfr-pass1/repo_git
bash scripts/submit_pyfr_pass5_diagnostic_gilbreth.sh

cd /Users/ashmitprabhu/github/nasa-hump2
bash scripts/fetch_gpu_pyfr_pass5_results.sh
MPLCONFIGDIR=docs/.mplconfig python3 scripts/postprocess_gpu_pyfr_pass5.py
'/Users/ashmitprabhu/Desktop/ParaView-6.1.0-RC1.app/Contents/bin/pvpython' \
scripts/render_gpu_pyfr_pass5_paraview.py velocity
bash scripts/build_report.sh
```

## 15 Conclusion

GPU PyFR PASS 5 closes the main ambiguity left by PASS 4. The reason  $C_p$  was unavailable is now explicit: the exported PASS 4 field itself is globally non-finite. PASS 5 therefore promotes a more robust workflow: audit first, extract only from finite fields, and rerun conservatively when the existing field is invalid. That is the right foundation for the first honest PyFR  $C_p$  comparison on the NASA hump.

## References